

Week 1 Workshop — Gradients & Curvature: From Topography to Hidden Fields

Geophysics IIIA (Potential Fields & Geothermics)

May 27, 2026

Purpose

Build intuitive and mathematical comfort with gradients (∇h) and curvature ($\nabla^2 h$) in 2-D, starting from topography, then transferring the concepts to hidden subsurface scalar fields (e.g., density, temperature). This prepares students for potential fields (Weeks 3–7) and diffusion/heat flow (Weeks 8–10). *Suggested pre-reading: Unit 1 notes on gradient, divergence, Laplacian, and Poisson/Laplace. (See Unit1.pdf.)* [1]

Learning outcomes

By the end of the workshop, students will be able to:

1. Draw and interpret the gradient vector field of a scalar map (magnitude and direction).
2. Explain what the Laplacian measures (local curvature) and where it is large.
3. Transfer gradient/curvature ideas from elevation to hidden fields; reason about subsurface contacts and sources/sinks.
4. Articulate why moving farther from sources produces smoother maps (“distance acts like a low-pass filter”).

Timing (180 min)

0–10 min Concept warm-up. Units of gradient; why a gradient is a vector (direction of steepest increase; magnitude = steepest slope).

10–65 min Hands-on gradients (maps). Using Mt. Shasta and Marble Canyon topographic maps, students draw gradient arrows (longer where steeper), then compare to computed arrows/contours. Discuss where $|\nabla h|$ is large/small and why. *(Based on gradient.pdf.)* [2]

65–90 min Mini-derivation & vocabulary. From slope to partial derivatives to ∇ ; introduce ∇^2 as divergence of ∇ (curvature). Keep algebra light; focus on meanings and units. *(See Unit1.pdf on grad/div/Laplacian.)* [1]

90–130 min Hidden fields page. Treat the map as bulk density $\rho(x, y)$ (or temperature T). Sketch $\nabla \rho$ arrows and mark likely subsurface boundaries (faults, intrusive contacts). For each boundary, state whether the property is sharp or gradational and what that implies for ∇^2 inside/outside sources.

130–165 min Sharp vs. smooth. Compare two maps: “close” vs “far” from a source. Identify visual cues (peak width, curvature, edge sharpness). Conclude: farther $\Rightarrow$ smoother; closer $\Rightarrow$ sharper.

165–180 min Debrief. How do these ideas appear in gravity/magnetics (continuation) and heat (diffusion)? Link to this week’s practical (explicit Laplacian smoothing on ETOPO2) and to PS1’s emphasis on units and assumptions. *(See PS1.)* [3]

Group structure

Groups of 3–4 with rotating roles each week: Analyst (math/units), Modeler (figures), Interpreter (geologic meaning), Critic (assumptions/uncertainty).

Materials

- Printed A3/A4 maps from the gradient worksheet (Mt. Shasta, Marble Canyon), blank overlays, fine markers. [2]
- One-page “Hidden Fields” insert (below) for sketching density/temperature gradients and marking likely subsurface contacts.

Hidden Fields insert (student page)

1) Gradient of a hidden scalar. Suppose the map shows shallow bulk density $\rho(x, y)$ instead of elevation. Draw arrows for $\nabla\rho$: longer where change is larger; direction toward increasing ρ . Label likely subsurface boundaries (faults, intrusive contacts, channel fills).

2) Boundary reasoning. For each boundary, is ρ piecewise constant (sharp contact) or gradational? What sign/magnitude of $\nabla^2\rho$ would you expect inside source regions vs outside (source-free) regions? (*Recall from Unit1: outside sources $\nabla^2\Phi = 0$; with sources $\nabla^2\Phi = -\rho/\kappa$ by analogy.*) [1]

3) Forward-model intuition. Sketch equipotential contours Φ and field lines $\mathbf{F} = -\nabla\Phi$ for a simple dense plug beneath a flat surface. Where would a surface gravimeter see the largest vertical gradient?

4) Heat-flow bridge. If the map showed temperature T , which regions suggest internal heat production? Relate to the sign of $\nabla^2 T$.

Instructor notes

Keep symbols/units consistent with the course notes; prioritize qualitative understanding. Emphasize that “farther $\Rightarrow$ smoother” returns in continuation (Weeks 6–7) and diffusion (Weeks 8–9). [1]

Further Reading:

- [1] Unit1.pdf: Course notes introducing work/potential, ∇ , ∇^2 , Poisson/Laplace. [?]]
- [2] gradient.pdf: Gradient worksheet with Mt. Shasta & Marble Canyon maps and solutions. [?]]
- [3] geophysics_ps1.pdf: Physical properties & back-of-the-envelope problem set. [?]]